Table 1 — Ablation Study: Effect of MoCLIP, S-VIB Gate, and Physics Smoother

Model Variant	Component Flags: MoCLIP	Component Flags: S-VIB ψ	Component Flags: Physics	Metrics: FGD ↓	Metrics: BC ↑	Metrics: L1 Vel ↓
SemTalk (baseline)	✗	✗	✗	0.4380	—	—
+ MoCLIP	✓	✗	✗	0.4263	—	—
+ MoCLIP + S-VIB	✓	✓	✗	—	— *pending*	—
+ MoCLIP + Phys. (arms only)	✓	✗	✓	—	— *pending*	—
+ MoCLIP + S-VIB + Phys. (arm)	✓	✓	✓	0.4180	—	—
+ MoCLIP + S-VIB + Phys. (full)	✓	✓	✓	**0.4137** ★	**0.7460**	**12.46**

✓/✗ = component enabled/disabled ★ = best result — = metric not yet evaluated amber rows = training run pending